

Program :

- '''- Download a new dataset related to greenskilling or sustainability*
- Import in python file*
- Note all the generalisations and analysis about the dataset, note relations of each feature/column to each other. (write it in text cells or comments)*
- if required do feature engineering*
- Handle null values according to your understanding of dataset*
- Used Seaborn library plots to visualise your dataset*
- Use scaling on numerical columns and pd.get_dummies / Label encoder on Categorical columns*
- Give your insights and answer why you did what you did.'''*

```
import pandas as p
from sklearn.preprocessing import *
import seaborn as s
import matplotlib.pyplot as plt
data = p.read_csv('data.csv')
print('Dataset Overview:')
data.info()
print('The person\'s age greater than 25 are:')
print(data[data['Age'] > 25])
print('Number of females in the dataset:')
print(data[data['Gender'] == 'Female'])
print('The peoples who belong to Ahmedabad city are:')
print(data[data['City'] == 'Ahmedabad'])
print('The number of null values in the dataset are:')
print(data.isnull().sum().sum())
if data.isnull().sum().sum() > 0:
    print('Handling null values by filling with mean for numerical columns and mode for
categorical columns.')
    for column in data.columns:
        if data[column].dtype == 'object':
            # Fill text columns with the most frequent value
            data[column] = data[column].fillna(data[column].mode()[0])
        else:
            # Fill numeric columns with the mean
            data[column] = data[column].fillna(data[column].mean())
print('Visualizing the dataset using Seaborn library:')
print('Scaling numerical columns and encoding categorical columns:')
print(minmax_scale(data.select_dtypes(include=['float64', 'int64'])))
print('Standard Scaled Data:')
print(StandardScaler().fit_transform(
    data.select_dtypes(include=['float64', 'int64'])))
print('Heatmap of the dataset:')
corr = data.corr(numeric_only=True)
s.heatmap(corr, annot=True)
plt.show()
```

```

print('Pairplot of the dataset:')
s.pairplot(data)
plt.show()
le = LabelEncoder()
data['Gender'] = le.fit_transform(data['Gender'])
data['City'] = le.fit_transform(data['City'])
data['Transport_Mode'] = le.fit_transform(data['Transport_Mode'])
data['Solar_Panels'] = le.fit_transform(data['Solar_Panels'])
data['Green_Certification'] = le.fit_transform(data['Green_Certification'])
data['Eco_Friendly'] = le.fit_transform(data['Eco_Friendly'])
print('Label Encoded Data:')
print(data)
print('Dummies for categorical columns:')
data = p.get_dummies(data, columns=[
    'Gender', 'City', 'Transport_Mode', 'Solar_Panels', 'Green_Certification', 'Eco_Friendly'])
print(data)
data.to_csv('processed_data.csv', index=False)
print('Insights:')
print('1. This dataset provides information about individual\'s daily lifestyle use.')
print('2. The dataset contains both numerical and categorical features.')
print('3. Most of the people choose car as their mode of transport, which is not eco-friendly.')
print('4. The age of the people in the dataset is mostly between 20-30 years, which indicates
that younger people are more conscious about their lifestyle choices.')

```

Output:

Dataset Overview:

```
<class 'pandas.core.frame.DataFrame'>
```

RangeIndex: 10 entries, 0 to 9

Data columns (total 13 columns):

#	Column	Non-Null Count	Dtype
0	Person_ID	10 non-null	int64
1	Age	10 non-null	int64
2	Gender	10 non-null	object
3	City	10 non-null	object
4	Transport_Mode	10 non-null	object
5	Electricity_Usage	9 non-null	float64
6	Water_Usage	9 non-null	float64
7	Waste_Recycled	9 non-null	float64
8	Solar_Panels	10 non-null	object
9	Green_Certification	9 non-null	object
10	Trees_Planted	9 non-null	float64
11	Sustainability_Score	10 non-null	int64
12	Eco_Friendly	10 non-null	object

dtypes: float64(4), int64(3), object(6)

memory usage: 1.1+ KB

The person's age greater than 25 are:

	Person_ID	Age	...	Sustainability_Score	Eco_Friendly
3	4	27	...	72	Yes
7	8	26	...	64	No
9	10	28	...	42	No

[3 rows x 13 columns]

Number of females in the dataset:

	Person_ID	Age	...	Sustainability_Score	Eco_Friendly
0	1	22	...	88	Yes
2	3	21	...	94	Yes
4	5	23	...	98	Yes
6	7	22	...	81	Yes
8	9	20	...	95	Yes

[5 rows x 13 columns]

The peoples who belong to Ahmedabad city are:

	Person_ID	Age	...	Sustainability_Score	Eco_Friendly
0	1	22	...	88	Yes
4	5	23	...	98	Yes
8	9	20	...	95	Yes

[3 rows x 13 columns]

The number of null values in the dataset are:

5

Handling null values by filling with mean for numerical columns and mode for categorical columns.

	Person_ID	Age	Gender	City	Transport_Mode	Electricity_Usage	...	Waste_Recycled	
	Solar_Panels	Green_Certification	Trees_Planted	Sustainability_Score	Eco_Friendly				
0	1	22	Female	Ahmedabad	Bus	185.0	...	75.000000	Yes
Yes	12.000000			88	Yes				
1	2	25	Male	Surat	Car	310.0	...	30.000000	No
No	2.000000			49	No				
2	3	21	Female	Vadodara	Bicycle	150.0	...	85.000000	Yes
Yes	15.000000			94	Yes				
3	4	27	Male	Rajkot	Metro	240.0	...	60.000000	No
Yes	8.000000			72	Yes				
4	5	23	Female	Ahmedabad	Walking	120.0	...	90.000000	Yes
Yes	20.000000			98	Yes				
5	6	24	Male	Surat	Car	215.0	...	40.000000	No
No	3.000000			55	No				
6	7	22	Female	Rajkot	Bus	200.0	...	68.000000	Yes
Yes	10.000000			81	Yes				
7	8	26	Male	Vadodara	Metro	260.0	...	62.333333	No
No	5.000000			64	No				

8	9	20	Female	Ahmedabad	Bicycle	140.0	...	88.000000	Yes
Yes	18.000000			95	Yes				
9	10	28	Male	Surat	Car	330.0	...	25.000000	No
No	10.333333			42	No				

[10 rows x 13 columns]

Visualizing the dataset using Seaborn library:

Scaling numerical columns and encoding categorical columns:

```
[[0.    0.25    0.30952381 0.21212121 0.76923077 0.55555556
  0.82142857]
 [0.11111111 0.625    0.9047619  0.81818182 0.07692308 0.
  0.125    ]
 [0.22222222 0.125    0.14285714 0.09090909 0.92307692 0.72222222
  0.92857143]
 [0.33333333 0.875    0.57142857 0.48484848 0.53846154 0.33333333
  0.53571429]
 [0.44444444 0.375    0.    0.    1.    1.
  1.    ]
 [0.55555556 0.5      0.45238095 0.6969697  0.23076923 0.05555556
  0.23214286]
 [0.66666667 0.25     0.38095238 0.43434343 0.66153846 0.44444444
  0.69642857]
 [0.77777778 0.75     0.66666667 0.57575758 0.57435897 0.16666667
  0.39285714]
 [0.88888889 0.       0.0952381  0.03030303 0.96923077 0.88888889
  0.94642857]
 [1.    1.    1.    1.    0.    0.46296296
  0.    ]]
```

Standard Scaled Data:

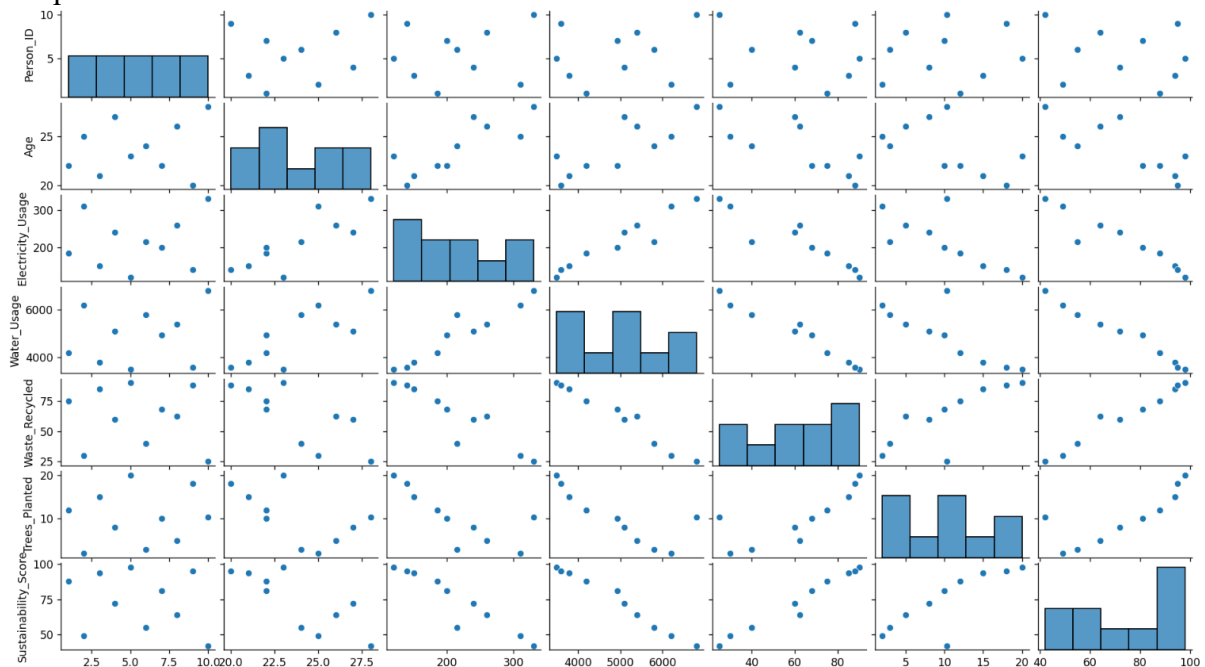
```
[[ -1.56669890e+00 -7.13746427e-01 -4.47711329e-01 -6.76810952e-01
   5.62324636e-01  2.88386892e-01  7.30794644e-01]
 [ -1.21854359e+00  4.75830951e-01  1.41775254e+00  1.16903710e+00
  -1.43540762e+00 -1.44193446e+00 -1.27631741e+00]
 [ -8.70388280e-01 -1.11027222e+00 -9.70041213e-01 -1.04598056e+00
   1.00626514e+00  8.07483297e-01  1.03958111e+00]
 [ -5.22232968e-01  1.26888254e+00  3.73092774e-01  1.53820671e-01
  -1.03586117e-01 -4.03741648e-01 -9.26359408e-02]
 [ -1.74077656e-01 -3.17220634e-01 -1.41775254e+00 -1.32285777e+00
   1.22823539e+00  1.67264397e+00  1.24543876e+00]
 [  1.74077656e-01  7.93051586e-02  0.00000000e+00  7.99867489e-01
  -9.91467121e-01 -1.26890232e+00 -9.67530937e-01]
 [  5.22232968e-01 -7.13746427e-01 -2.23855665e-01  0.00000000e+00
   2.51566284e-01 -5.76773784e-02  3.70543763e-01]
 [  8.70388280e-01  8.72356744e-01  6.71566994e-01  4.30697878e-01
   0.00000000e+00 -9.22838054e-01 -5.04351233e-01]
 [  1.21854359e+00 -1.50679801e+00 -1.11927832e+00 -1.23056537e+00
```

1.13944729e+00 1.32657970e+00 1.09104552e+00]
[1.56669890e+00 1.66540833e+00 1.71622676e+00 1.72279151e+00
-1.65737787e+00 3.07366817e-16 -1.63656829e+00]]

Heatmap of the dataset:



Pairplot of the dataset:



Label Encoded Data:

	Person_ID	Age	...	Sustainability_Score	Eco_Friendly
0	1	22	...	88	1
1	2	25	...	49	0

2	3	21	...	94	1
3	4	27	...	72	1
4	5	23	...	98	1
5	6	24	...	55	0
6	7	22	...	81	1
7	8	26	...	64	0
8	9	20	...	95	1
9	10	28	...	42	0

[10 rows x 13 columns]

Dummies for categorical columns:

	Person_ID	Age	...	Eco_Friendly_0	Eco_Friendly_1
0	1	22	...	False	True
1	2	25	...	True	False
2	3	21	...	False	True
3	4	27	...	False	True
4	5	23	...	False	True
5	6	24	...	True	False
6	7	22	...	False	True
7	8	26	...	True	False
8	9	20	...	False	True
9	10	28	...	True	False

[10 rows x 24 columns]

Insights:

1. This dataset provides information about individual's daily lifestyle use.
2. The dataset contains both numerical and categorical features.
3. Most of the people choose car as their mode of transport, which is not eco-friendly.
4. The age of the people in the dataset is mostly between 20-30 years, which indicates that younger people are more conscious about their lifestyle choices.